Wide-Angle Multislice Propagation for Electron Microscopy: Comparison of Fresnel, Angular-Spectrum, and Wave-Propagation Methods

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Abstract

Conventional Fresnel multislice (F-MS) is the standard method for modelling electron–specimen interactions in transmission electron microscopy (TEM) due to its computational efficiency and ability to simulate both perfect and defective crystals. However, its paraxial propagation kernel loses accuracy when large scattering angles contribute to the exit wave in thick samples. Here we compare slice-based fast Fourier transform (FFT) propagation methods that preserve the practical multislice workflow while improving wide-angle propagation. We introduce wave-propagation multislice (WP-MS), an adaptation of the wave propagation method (WPM) from scalar light optics, and compare it with F-MS and angular-spectrum multislice (AS-MS). The three methods are first benchmarked against an ODE solution of the second-order scalar Klein–Gordon equation on the same transverse grid for a representative Au [100] specimen. We then compare their predictions for convergent-beam electron diffraction (CBED) in Si [111] and Au [100]. AS-MS removes most of the wide-angle error of F-MS at essentially the same per-slice computational cost. WP-MS supplies a further local-medium correction at higher cost, but its difference from AS-MS is smaller than the F-MS discrepancy under the high-energy conditions studied here.

Keywords: electron microscopy, multislice, electron diffraction, Fresnel propagation, angular spectrum, wave propagation method

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Glossary

AS-MS Angular-spectrum multislice.

BPM Beam-propagation method.

CBED Convergent-beam electron diffraction.

FC-MS Fully corrected multislice.

FFT Fast Fourier transform.

F-MS Fresnel multislice.

KG Klein–Gordon.

ODE Ordinary differential equation.

STEM Scanning transmission electron microscopy.

TEM Transmission electron microscopy.

WPM Wave propagation method.

WP-MS Wave-propagation multislice.

1. Introduction

The traditional image-simulation workflow in electron microscopy assumes a structural model, calculates an image from that model, and compares the result qualitatively with experiment by trial and error. In modern reconstruction workflows, including ptychography and related phase-retrieval methods, the recorded data are instead used to recover a projected potential, phase shift, or three-dimensional structure. The electron–matter interaction model is therefore part of the reconstruction algorithm itself, so the accuracy of the forward model directly affects the recovered structure.

The standard forward model in electron microscopy is paraxial Fresnel multislice (F-MS) [1, 2]. F-MS is simple, fast, and well suited to conventional high-energy TEM and scanning TEM (STEM) simulations of thin, low-Z specimens. However, for low incident energies, thick specimens, high-Z materials, or simulations that retain high-angle diffraction information, F-MS can introduce significant errors because its propagation kernel is paraxial and because standard forward multislice neglects backscattering [3–6].

The smallest correction to F-MS is to replace the parabolic Fresnel propagator between slices by an exact spherical propagator. To our knowledge, Ming and Chen [6] were the first to propose this full-spherical-propagator correction explicitly in electron microscopy. We refer to this method as angular-spectrum multislice (AS-MS), following the terminology used in scalar light optics [7]. AS-MS keeps the same split-step structure and the same FFT-based cost as F-MS, but restores the exact free-space dispersion relation. It is therefore a minimal wide-angle correction, although it still treats the specimen as a sequence of projected phase gratings separated by vacuum propagation.

More complete wide-angle methods seek to improve the propagation inside the specimen, or to use the full second-order wave equation as a reference. The wave equation can be separated into forward- and backward-propagating branches, and in the forward branch, longitudinal propagation is governed by a square-root operator that contains both the specimen potential and the transverse Laplacian. The Chen–Van Dyck method, also known as fully corrected multislice (FC-MS), expands this operator in a Taylor series and retains higher-order terms that couple the specimen potential to transverse derivatives [6, 8]. This improves the treatment of large scattering angles, low-energy electrons, thick specimens, and high-Z materials, and in its more complete form gives access to backward-propagating components. The cost is that the square-root operator is no longer a simple Fourier-space phase factor, and the calculation can be expensive depending on the number of terms retained in the expansion.

A closely related method was already developed earlier in scalar light optics, known as Padé beam-propagation methods which approximate the same square-root operator by rational functions rather than Taylor polynomials [9]. These approximants can represent the square-root dispersion relation over a wider angular range than a truncated Taylor series of the same order, but they still introduce substantial overhead relative to F-MS or AS-MS. Finally, direct integration of the full second-order wave equation as an ordinary differential equation (ODE) provides a more accurate numerical reference, and has been used to assess paraxial and corrected multislice methods [3, 5]. These studies indicate that a dominant high-angle error of F-MS arises from the parabolic propagation step. Hillier et al. [4] also showed that retaining the second longitudinal derivative is important for the positions of Laue-zone rings in low-energy Si CBED patterns. However, ODE and higher-order corrected methods are too expensive to use directly as routine forward models in large-scale inverse reconstructions.

To obtain accurate solutions of the time-independent wave equation without losing the efficiency of FFT-based propagation, the light-optics community has developed alternatives that avoid choosing a Taylor or Padé expansion order. One such method is the wave propagation method (WPM), introduced by Brenner and Singer [10]. WPM treats the medium as locally homogeneous over small transverse regions, applies homogeneous angular-spectrum propagators for representative refractive indices, and recombines the resulting fields according to the local refractive-index distribution. This retains a potential-dependent wavelength within the propagation step without explicitly forming the full dense square-root operator. Efficient and differentiable variants have been developed in the scalar-optics literature [11, 12]. The method is therefore a natural starting point for a practical wide-angle multislice algorithm in electron microscopy, but to our knowledge it has not yet been systematically assessed in this context - see table 1 which summarises the related developments in light and electron optics for the method families compared here.

Method family	**Light-optics reference**	**Electron-optics reference**	**Description**
Fresnel MS	Fresnel (1818); Fleck–Morris–Feit (1976) [13, 14]	Cowley–Moodie (1957); Ishizuka–Uyeda (1977) [1, 2]	Split-step propagation with the paraxial Fresnel kernel.
Angular Spectrum MS	Booker–Clemmow (1950); Korpel et al. (1986) [15, 16]	Ming–Chen (2013) [6]	Same split-step structure, but with the exact free-space propagator.
Padé/series-expansion BPMs	Hadley (1992); Lee–Voges (1994) [9, 17]	Chen–Van Dyck (1997) [8]	Rational or Taylor-series approximations to the square-root propagation operator.
Wave Propagation Method	Brenner–Singer (1993); Schmidt et al. (2016) [10, 11]	This work	Local homogeneous angular-spectrum kernels applied inside the medium.
ODE	Banerjee–Sharma (1989) [18, 19]	Rother–Scheerschmidt (2009)*; Wacker–Schröder (2015) [3, 5]	Reciprocal-space and direct references for non-paraxial propagation.

*Rother–Scheerschmidt used a forward-scattering approximation in Fourier space for a periodic crystal, rather than direct real-space integration of the full second-order ODE.

Table 1: Historical correspondence between the propagation methods used here and closely related developments in light and electron optics.

The aim of this work is to adapt and modify the WPM from the light optics community (which we call Wave Propagation Multislice, WP-MS) and assess it's benefits in the context of electron microscopy. We initially compare

WP-MS, F-MS, and AS-MS with a forward ODE reference for Au [100], and then compare the three multislice methods in representative Si [111] and Au [100] CBED calculations. The comparison isolates the dominant correction to F-MS and tests whether the local-wavelength treatment in WP-MS produces a numerically resolved additional change under the conditions considered here.

2. Wave-Equation Models and Propagation Algorithms

2.1. Wave equation and refractive index

Throughout this work we use spatial frequencies in cycles per unit length. The vacuum spatial frequency is therefore

$$k_0 = \frac{1}{\lambda}, \tag{1}$$

where λ is the relativistic electron wavelength. We use the Fourier transform convention

$$\mathcal{F}\{f(x,y)\}(k_x,k_y) = \int f(x,y)\exp\left[-2\pi i(k_x x + k_y y)\right]\,dx\,dy, \tag{2}$$

and write $k_\perp^2 = k_x^2 + k_y^2$.

For high-energy elastic scattering in an electrostatic potential, we start from the scalar relativistically corrected Schrödinger equation [20, 21]

$$\left[-\frac{\hbar^2}{2\gamma m_0}\nabla^2 + E_{\mathrm{pot}}(\mathbf{r})\right]\psi(\mathbf{r}) = \frac{p_0^2}{2\gamma m_0}\psi(\mathbf{r}), \tag{3}$$

where ψ is the spatial wavefunction, m_0 is the electron rest mass, γ is the Lorentz factor, and $p_0 = h/\lambda = hk_0$ is the vacuum momentum. We write the electrostatic potential in volts as $V(\mathbf{r})$ and the corresponding electron potential energy as $E_{\mathrm{pot}}(\mathbf{r}) = -eV(\mathbf{r})$, with $e > 0$.

Multiplying equation (3) by $-2\gamma m_0/\hbar^2$ gives a Helmholtz equation for the same wavefunction,

$$\nabla^2\psi + (2\pi k_0)^2 n_{\mathrm{S}}^2(\mathbf{r})\psi = 0, \tag{4}$$

where the Schrödinger refractive index is

$$n_{\mathrm{S}}^2(\mathbf{r}) = 1 - \frac{2\gamma m_0}{h^2 k_0^2}E_{\mathrm{pot}}(\mathbf{r}) = 1 + \frac{\sigma}{\pi k_0}V(\mathbf{r}), \tag{5}$$

and the conventional TEM interaction constant is

$$\sigma = \frac{2\pi\gamma m_0 e}{h^2 k_0} = \frac{2\pi\gamma m_0 e\lambda}{h^2}. \tag{6}$$

The electron-optical refractive-index description has a long history [22–24], and the refractive index notation is useful because the choice of scalar wave equation can be expressed through the choice of $n(\mathbf{r})$, separating approximations to the wave equation from approximations to the potential. Letting $E_0 = m_0c^2$ denote the electron rest energy, the time-independent Klein–Gordon (KG) equation gives

$$n_{\mathrm{KG}}^2(\mathbf{r}) = \frac{\left[E_{\mathrm{tot}} - E_{\mathrm{pot}}(\mathbf{r})\right]^2 - E_0^2}{E_{\mathrm{tot}}^2 - E_0^2}. \tag{7}$$

Here E_{tot} is the incident-electron energy in vacuum, including the rest energy, so for an electron accelerated through a voltage U, $E_{\mathrm{tot}} = E_0 + eU = \gamma E_0$.

The relativistically corrected Schrödinger refractive index is a linearised potential approximation to the Klein–Gordon refractive index while retaining the relativistic vacuum momentum through k_0. Therefore, changing from n_{S} to n_{KG} changes the refractive-index model, whereas the choice of propagation method (see Section 2) changes how a given input field is propagated through the refractive-index distribution.

2.2. Fresnel and angular-spectrum multislice

To connect Equation (4) to conventional multislice, we make the standard high-energy (slowly varying-envelope) approximation used in electron microscopy [21]. Writing the wavefunction as a rapidly oscillating carrier multiplied by a slowly varying envelope,

$$\psi(x, y, z) = \exp(2\pi i k_0 z)\tilde{\psi}(x, y, z), \tag{8}$$

gives the parabolic wave equation

$$\frac{\partial\tilde{\psi}}{\partial z} = \frac{i}{4\pi k_0}\nabla_\perp^2\tilde{\psi} + i\pi k_0\left[n^2(x, y, z) - 1\right]\tilde{\psi}. \tag{9}$$

For a chosen refractive-index model n, the material substep across slice j is

$$t_j^{(n)}(x, y) = \exp\left[i\pi k_0\int_{z_j}^{z_j+\Delta z}\left(n^2(x, y, z) - 1\right)\,dz\right]. \tag{10}$$

For the corrected Schrödinger index in Equation (5), this reduces exactly to the conventional TEM transmission function

$$t_j^{(\mathrm{S})}(x,y) = \exp\left[i\sigma V_j(x,y)\right], \qquad V_j(x,y) = \int_{z_j}^{z_j+\Delta z} V(x,y,z)\,dz. \tag{11}$$

In all numerical comparisons below, F-MS and AS-MS use Equation (10) with $n = n_{\mathrm{KG}}$, so their transmission, the WP-MS propagator, and the ODE reference are constructed from the same refractive-index model and potential stack.

Writing the selected transmission function simply as t_j, Fresnel multislice (F-MS) applies it and then propagates the wave through vacuum with the paraxial Fresnel kernel,

$$P_{\mathrm{F}}(k_x,k_y;\Delta z) = \exp\left(-i\pi\frac{k_\perp^2}{k_0}\Delta z\right), \qquad k_\perp^2 = k_x^2 + k_y^2. \tag{12}$$

Dropping the tilde for notational simplicity, one slice can be written as

$$\psi_{j+1}(x,y) \approx \mathcal{F}^{-1}\left[P_{\mathrm{F}}(k_x,k_y;\Delta z)\mathcal{F}\{t_j(x,y)\psi_j(x,y)\}\right]. \tag{13}$$

Computing the above equation is fast because each slice requires only one FFT pair per slice. Its free-space approximation is the expansion of the longitudinal spatial frequency as

$$k_z = \sqrt{k_0^2 - k_\perp^2} \simeq k_0 - \frac{k_\perp^2}{2k_0}.$$

After removal of the carrier phase $\exp(2\pi i k_0 z)$, this expansion gives the Fresnel kernel in equation (12). Conventional multislice is therefore both a split-step approximation to the specimen and a small-angle approximation to the longitudinal dispersion relation.

The angular spectrum method keeps the same split-step structure but replaces the Fresnel kernel by the exact free-space dispersion relation:

$$\psi_{j+1}(x,y) \approx \mathcal{F}^{-1}\left[\exp\left(2\pi i\left[\sqrt{k_0^2 - k_\perp^2} - k_0\right]\Delta z\right)\mathcal{F}\{t_j(x,y)\psi_j(x,y)\}\right]. \tag{14}$$

Apart from the change in the Fourier-space phase factor, the algorithm is the same as standard multislice and has the same $\mathcal{O}(P\log P)$ per-slice cost, where P is the number of transverse pixels. It is therefore the smallest change one can make to F-MS in order to restore the exact free-space dispersion relation. The approximation that remains is the split-step treatment of the specimen: the potential is still applied as a thin transmission function, while propagation between slices is still performed in vacuum.

2.3. Classical wave propagation method in scalar optics

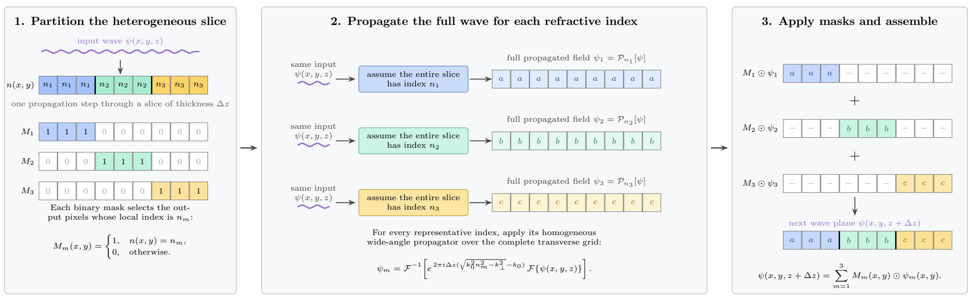

Figure 1: Classical wave propagation method (WPM). The schematic follows the algorithm of Schmidt et al. [11]. The input wave is propagated using a bank of homogeneous angular-spectrum kernels evaluated at representative refractive indices n_m. At each output pixel, a mask selects the propagated field associated with the local refractive index. The WP-MS adaptation introduced below replaces this hard selection by a smooth two-bin interpolation (Figure 2).

The Wave Propagation Method (WPM) removes the vacuum-propagation approximation in AS-MS and instead includes the refractive-index dependence of the wavelength in the propagation kernel. In this sense, the local interaction with the medium is incorporated into a single propagation update, rather than being applied as a separate transmission function. Returning to the full second-order Helmholtz equation (4), the locally z-invariant problem can be written as an operator equation with forward and backward branches [11]. Ignoring the backward branch gives

$$\partial_z \psi = 2\pi i \left[\sqrt{k_0^2 n^2(x, y, z) + \frac{1}{4\pi^2} \nabla_\perp^2} - k_0 \right] \psi. \tag{15}$$

The central concern is that the square-root operator inside the wave equation mixes the potential and the transverse derivative, and cannot be evaluated as a single Fourier-space propagator. Padé BPM and FC-MS approximate this operator using rational or series expansions of the square-root term, which do not reduce to a single FFT-diagonal phase factor.

The AS-MS propagation kernel in Equation (14) contains

$$\left[\sqrt{k_0^2 - k_\perp^2} - k_0\right]$$

but not the local refractive index. WPM introduces this dependence locally by replacing k_0 inside the square root by $k_0 n(x, y)$, giving

$$\left[\sqrt{k_0^2 n^2(x, y) - k_\perp^2} - k_0\right].$$

For a single refractive-index value, this kernel can be evaluated straightforwardly with the angular spectrum method. The traditional WPM as introduced by Brenner and Singer simply extends the angular spectrum method by evaluating a bank of homogeneous angular-spectrum propagators for each refractive index in the slice, and then retaining the propagated field corresponding to the local refractive index at each output pixel.

This operation is obviously simplistic, because output pixels can receive a propagated field that travelled from some pixels with the incorrect refractive index. However, the approximation lives in the fact that those different refractive indices are further away from the local refractive-index value of that pixel, and therefore contribute less to the final result. In this manner, the propagation step accounts for the local wavelength inside the specimen without explicitly evaluating the full square-root operator. Schmidt et al. [11] provided a more rigorous derivation motivating the WPM, and showed that the leading error from neglecting coupling between neighbouring refractive-index regions affects amplitude rather than phase; consequently, the WPM phase propagation is correct up to second order.

Figure 1 shows the same local-homogeneous idea in the form used by our electron-microscopy adaptation below. In the classical WPM as implemented by Schmidt et al., a mask separates each unique refractive index in a slice, the angular-spectrum kernel is calculated for each index, and the mask selects the corresponding solution from the bank of propagated fields. Schmidt et al. [11] made the method practical for many optical simulations by grouping pixels with the same refractive index, as only one homogeneous angular-spectrum propagation is then required for each distinct index value. If a slice contains M such values, the cost is reduced to approximately

$$\mathcal{O}(MP \log P),$$

instead of a direct all-to-all evaluation.

This saving is large in light-optical simulations where a slice may contain only a small number of unique refractive indices. It is less immediate in electron microscopy, where the electrostatic potential varies continuously and almost every pixel may have a different refractive index. A second issue is that masking introduces sharp masks, which are inconvenient for gradient-based optimisation because the assignment of pixels to refractive-index bins is not smoothly differentiable.

2.4. Wave propagation multislice

We refer to our electron-microscopy adaptation of WPM as wave propagation multislice (WP-MS), to distinguish it from the classical WPM used in scalar light optics. The main practical issue is that a typical sample slice used for electron-microscopy simulations has a nearly continuous refractive-index distribution, so grouping exactly by refractive-index value would require almost one homogeneous propagation per pixel. We therefore use a differentiable binned version of WPM, partly following the soft-partition idea of [12], but generalised from two refractive-index values to many reference values.

For each slice, we first discretise the refractive-index range $[n_{\min}, n_{\max}]$ into N_{bins} discrete reference values, denoted by $\{n_{\text{ref}}^{(k)}\}_{k=1}^{N_{\text{bins}}}$.

Most pixels in a typical slice are close to vacuum, with $n \simeq 1$, whereas the larger refractive-index values associated with atoms occupy a smaller fraction of the slice. We therefore use non-uniform reference values rather than linearly spaced bins, allocating more bins near $n = 1$. The reference indices are distributed according to

$$n_{\text{ref}}^{(k)} = n_{\min} + (n_{\max} - n_{\min}) \left(\frac{k-1}{N_{\text{bins}} - 1} \right)^{\gamma}, \quad k = 1, \ldots, N_{\text{bins}}, \tag{16}$$

with $\gamma = 2$ in the simulations below. Values $\gamma > 1$ give denser binning near the vacuum refractive index. Figure 2 illustrates the binning algorithm.

Next, for a propagation step Δz, we compute a bank of propagated fields $u_k(x, y)$. Each field u_k corresponds to the diffraction of the input field $u_{\text{in}}(x, y)$ assuming a homogeneous medium with refractive index $n_{\text{ref}}^{(k)}$. This is performed efficiently using batch FFT operations with the angular spectrum kernel:

$$u_k(x,y) = \mathcal{F}^{-1}\left[\mathcal{F}\{u_{\text{in}}\}\exp\left(2\pi i\Delta z\left[\sqrt{(k_0 n_{\text{ref}}^{(k)})^2 - k_x^2 - k_y^2} - k_0\right]\right)\right]. \quad (17)$$

At each position (x,y), we find the two reference indices on either side of the local refractive index $n(x,y)$: $n_L = n_{\text{ref}}^{(k)}$ and $n_R = n_{\text{ref}}^{(k+1)}$, such that $n_L \leq n(x,y) \leq n_R$. We adopt a differentiable interpolation scheme by defining a local interpolation weight

$$w_{\text{raw}} = \frac{n(x,y) - n_L}{n_R - n_L}, \quad w_{\text{raw}} \in [0,1]. \quad (18)$$

To obtain smooth gradients at the bin boundaries, we map this linear weight through the cubic smoothstep function

$$w = S(w_{\text{raw}}) = 3w_{\text{raw}}^2 - 2w_{\text{raw}}^3. \quad (19)$$

The smoothstep derivative vanishes at both ends of each interval, avoiding the gradient discontinuity of piecewise-linear interpolation at a bin boundary.

The final field at the next step $u(x,y,z+\Delta z)$ is computed as the weighted superposition of the fields propagated at the bounding indices:

$$u(x,y) = (1-w)u_L(x,y) + wu_R(x,y). \quad (20)$$

The local refractive index first determines the two bounding reference indices n_L and n_R, and the weight w specifies the interpolation coordinate after the smoothstep mapping. The fields u_L and u_R are the full propagated fields obtained by assuming homogeneous media with refractive indices n_L and n_R, respectively, and the output pixel is formed from their local two-bin mixture.

Equivalently, the WP-MS update can be written as

$$\psi(x,y,z+\Delta z) \approx \sum_{m=1}^{N_{\text{bins}}} w_m(x,y)\mathcal{F}^{-1}\Big[\exp\Big(2\pi i\Big[\sqrt{k_0^2(n_{\text{ref}}^{(m)})^2 - k_\perp^2} - k_0\Big]\Delta z\Big) \times \mathcal{F}\{\psi(x,y,z)\}\Big]. \quad (21)$$

For a pixel bracketed by bins k and $k+1$, the weights in Equation (21) are $w_k = 1 - w$, $w_{k+1} = w$, and zero for all other bins. This gives an approximation to the one-way medium propagator in equation (15). For a homogeneous slice, the bank collapses to a single homogeneous angular-spectrum propagation. For an inhomogeneous specimen, the method is more expensive than AS-MS because many FFT-based propagations are required per slice, but it accounts for the local wavelength inside the specimen rather than propagating every slice with the vacuum wavelength.

Because the binning and interpolation are smooth, the update is differentiable with respect to the refractive-index map $n(x, y)$ and can be used in automatic differentiation workflows. The cost is set by the number of bins because each propagation step requires N_{bins} FFT-based propagations rather than one propagation as in AS-MS. This is still much cheaper than evaluating a separate propagation for every distinct pixel value. The accuracy therefore depends on whether the chosen bins resolve the refractive-index distribution of the slice. The relevant convergence test is to increase N_{bins} and compare the propagated field over the complete specimen thickness until the field error is within an acceptable tolerance.

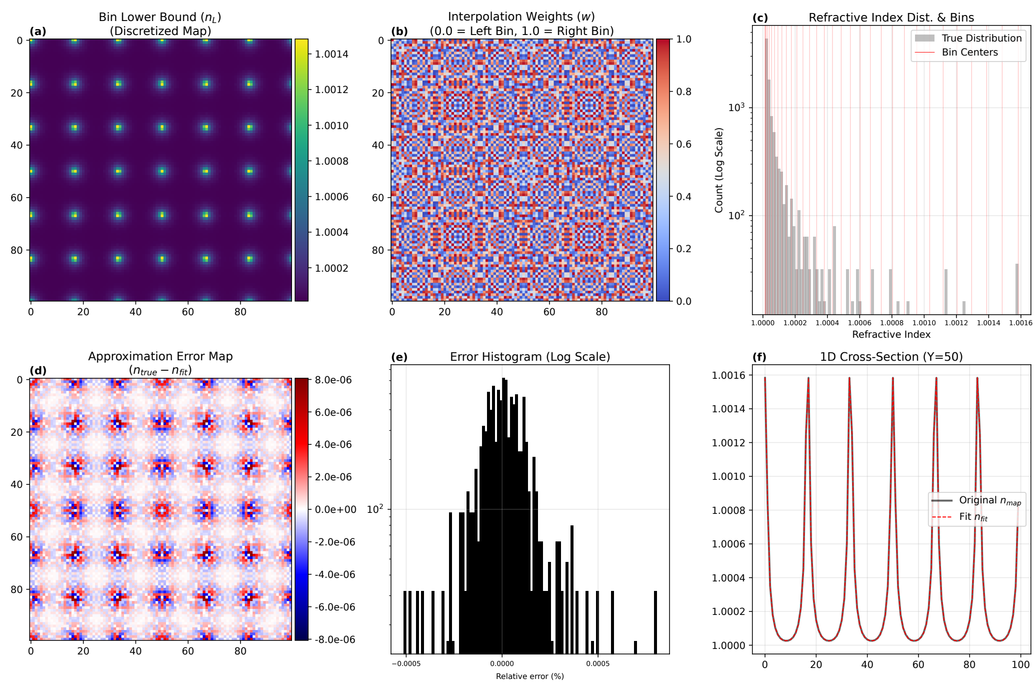

Figure 2: Diagnostics of the automatic-differentiation-compatible WP-MS binning strategy for a 200 keV Au [100] calculation using one 4.08 Å projected slice and 32 reference indices. (a) Lower-bin map n_L, showing the lower refractive-index bound assigned to each pixel. (b) Smooth interpolation weight w, where low values favour the lower-bin propagated field u_L and high values favour the upper-bin propagated field u_R. (c) Histogram of the true refractive-index distribution with the reference-bin positions superimposed. (d) Spatial approximation error $n_{\text{true}} - n_{\text{fit}}$. (e) Histogram of the fitting error with a logarithmic count axis. (f) One-dimensional cross-section comparing the original refractive-index profile with the interpolated profile n_{fit}.

Figure 2 illustrates how the two-bin interpolation scheme is applied to a representative slice. Panels (a) and (b) separate the binning construction into the interval assignment and the interpolation coordinate. In panel (a), the lower-bin map identifies the reference interval used at each pixel, whereas panel (b) gives the position of the true refractive index inside that interval through the interpolation weight. The diagnostic refractive-index fit, $n_{\text{fit}} = (1 - w)n_L + wn_R$, uses the same two-bin interpolation as the field update. Panels (c), (e), and (f) illustrate the index fit for this slice, while panel (d) shows that its largest residuals occur around steep atomic peaks. This diagnostic shows where the interpolation is largest; it is not a field-accuracy or full-thickness convergence test.

In practice, WP-MS is most useful with GPU acceleration. We used $N_{\text{bins}} = 256$ for the Au [100] ODE benchmark and $N_{\text{bins}} = 128$ for the Si [111] CBED calculation and 64 for the Au [100] CBED calculation. The diagnostic in Figure 2 uses 32 reference indices.

2.5. Forward ODE reference

We use an ordinary differential equation (ODE) reference solution to create a ground truth for the Au [100] benchmark. The ODE reference is a direct numerical solution of the second-order Klein–Gordon Helmholtz equation in Equation (4), with the same refractive-index model as the multislice methods. The transverse coordinates are represented on the same laterally periodic FFT grid used by the multislice methods, while z is left as the continuous evolution variable. The transverse pixel size and field of view therefore set the retained transverse spatial frequencies, or equivalently the largest scattering angles represented on the grid. The method does not attempt to resolve the rapidly oscillating electron wavelength in all three spatial directions on a Cartesian mesh; instead, the adaptive ODE integration resolves the rapid longitudinal oscillation along z.

Substituting n_{KG} into the Helmholtz equation gives

$$\partial_z^2 \psi = -\left[\nabla_\perp^2 + (2\pi k_0)^2 n_{\text{KG}}^2(x, y, z)\right] \psi. \tag{22}$$

Introducing $\phi = \partial_z \psi$ gives the equivalent first-order system

$$\frac{d}{dz}\begin{pmatrix}\psi \\ \phi\end{pmatrix} = \begin{pmatrix}\phi \\ -\left[\nabla_\perp^2 + (2\pi k_0)^2 n_{\text{KG}}^2(x, y, z)\right]\psi\end{pmatrix}. \tag{23}$$

We integrate this system with an adaptive high-order Runge–Kutta method, using the Dopri8 scheme with relative and absolute tolerances of 10^{-8} and 10^{-10}, respectively. The maximum internal step size is set from the largest retained discrete frequency. The transverse Laplacian is evaluated spectrally using FFTs, while the n_{KG}^2 term is applied by pointwise multiplication in real space. The discrete potential supplied to the solver is interpreted as the slice-averaged electrostatic potential, so n_{KG}^2 is constant in z within each slice and discontinuous only at slice boundaries. The adaptive step-size controller is clipped at those boundaries, so no accepted ODE step crosses a jump in the right-hand side. Thus, for a fixed transverse grid and slice stack, the ODE reference solves the corresponding periodic FFT-grid Klein–Gordon problem to the specified ODE tolerance.

To initialise the calculation on the forward-propagating vacuum branch, the entrance wave $\psi(z = 0)$ is supplemented by the longitudinal derivative

$$\phi(z = 0) = \mathcal{F}^{-1}\left[2\pi i\sqrt{k_0^2 - k_\perp^2}\,\mathcal{F}\{\psi(z = 0)\}\right], \tag{24}$$

where the square-root branch is chosen to give the forward-propagating vacuum solution, and the decaying branch for any evanescent Fourier components on the grid.

With this reference defined, the three multislice methods and the ODE benchmark can be seen as different approximations within the same scalar Helmholtz wave equation framework, once the refractive-index model is fixed. In the ODE benchmark below, all methods use the same per-slice n_{KG} map.

F-MS uses a split-step specimen approximation together with the paraxial free-space propagator, and AS-MS keeps the same split-step structure but restores the exact free-space spherical propagator. WP-MS remains one-way, but it does not use a separate transmission step; instead, it incorporates the refractive-index dependence into a local-homogeneous medium propagation update. By contrast, the ODE reference keeps the second-order Klein–Gordon Helmholtz equation after transverse discretisation and evolves the transverse Laplacian and specimen potential simultaneously, which is why it is used as the reference calculation in the first numerical experiment.

2.6. Numerical implementation

All specimen potentials were generated with abTEM 1.0.9 [25], using finite projection and the Lobato–Van Dyck independent-atom parametrisation [26]. The calculations are static, elastic, and use no thermal displacements.

3. Benchmarks and Results

We use three numerical experiments to benchmark the methods against an ODE reference and against each other. The first experiment adapts the geometry and tracked beams of the Au [100] benchmark of Rother and Scheerschmidt [3], so that F-MS, AS-MS, and WP-MS can be compared with a second-order scalar Klein–Gordon ODE solution. The ODE reference is not used in the remaining tests because of its high computational cost. Instead, the second experiment compares the FFT-based methods in a lower-Z Si [111] CBED calculation, and the third uses a thick Au [100] convergent-probe calculation.

3.1. Au [100] ODE Benchmark

To benchmark the implementation, we used the Au [100] geometry of the simulation presented in Figure 3 of Rother and Scheerschmidt [3]. A plane wave at 300 keV was propagated through a static face-centred cubic Au crystal with lattice constant $a = 4.08$ Å. Every method received the same slice-averaged electrostatic potential, generated using the Lobato parametrisation with finite projection on a 128×128 transverse grid. Each unit cell was divided into 256 slices, giving $\Delta x = \Delta y = 0.031875$ Å and $\Delta z = 0.0159375$ Å. F-MS and AS-MS used Equation (10) with $n = n_{\mathrm{KG}}$; WP-MS used the same n_{KG} in its local-medium propagator; and the ODE used the corresponding n_{KG}^2.

Periodic lateral boundary conditions are implicit because each method uses Fourier propagation, so the crystal is repeated in the transverse directions. The ODE was integrated with Dopri8 using relative and absolute tolerances of 10^{-8} and 10^{-10}, respectively. To test a thicker object than that of Rother and Scheerschmidt [3], we extended the 25-unit-cell ($\sim$ 10 nm) range shown in the original benchmark to 100 unit cells ($\sim$ 40 nm).

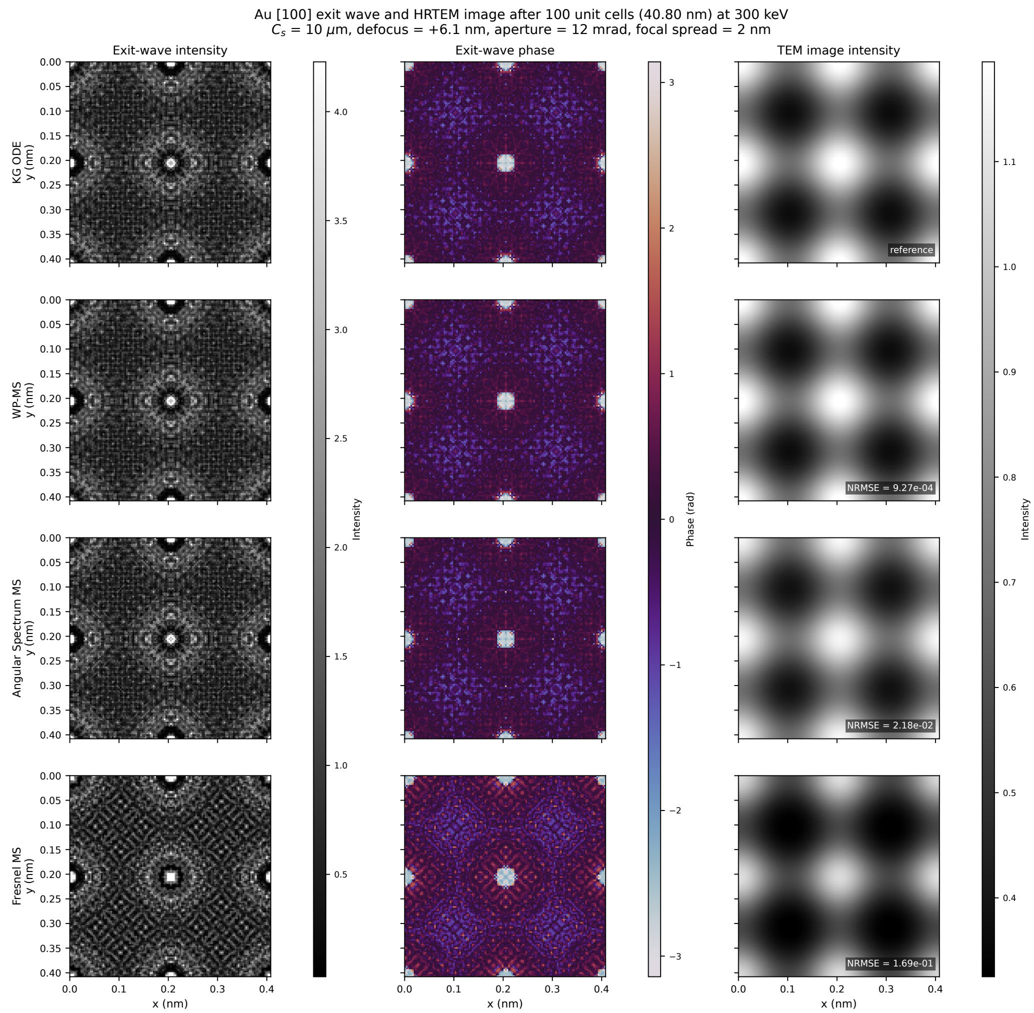

Figure 3: Au [100] ODE benchmark after 100 unit cells (40.8 nm). The first two columns show the real-space exit-wave intensity and phase, and the rightmost column shows the high-resolution TEM image intensity after applying a representative aberration-corrected 300 keV transfer function with residual $C_s = 10\ \mu$m, defocus of +6.0 nm, a 12 mrad objective aperture, and 2 nm focal spread. An independent global phase is removed from each exit wave before plotting the phase. Colour scales are shared across methods within each column, and the image-plane normalised root-mean-square differences are defined as $\|I - I_{\mathrm{ODE}}\|_2 / \|I_{\mathrm{ODE}}\|_2$.

Figure 3 shows the exit-wave intensity, phase, and transfer-function image intensity after propagation through 100 unit cells (40.8 nm) of Au [100]. The F-MS exit-wave intensity and phase are qualitatively different from AS-MS, WP-MS, and the Klein–Gordon ODE reference, whereas the latter three remain visually similar. After the microscope transfer function is applied, however, the image-plane differences are much less apparent because the objective aperture and transfer function suppress part of the high-angle information.

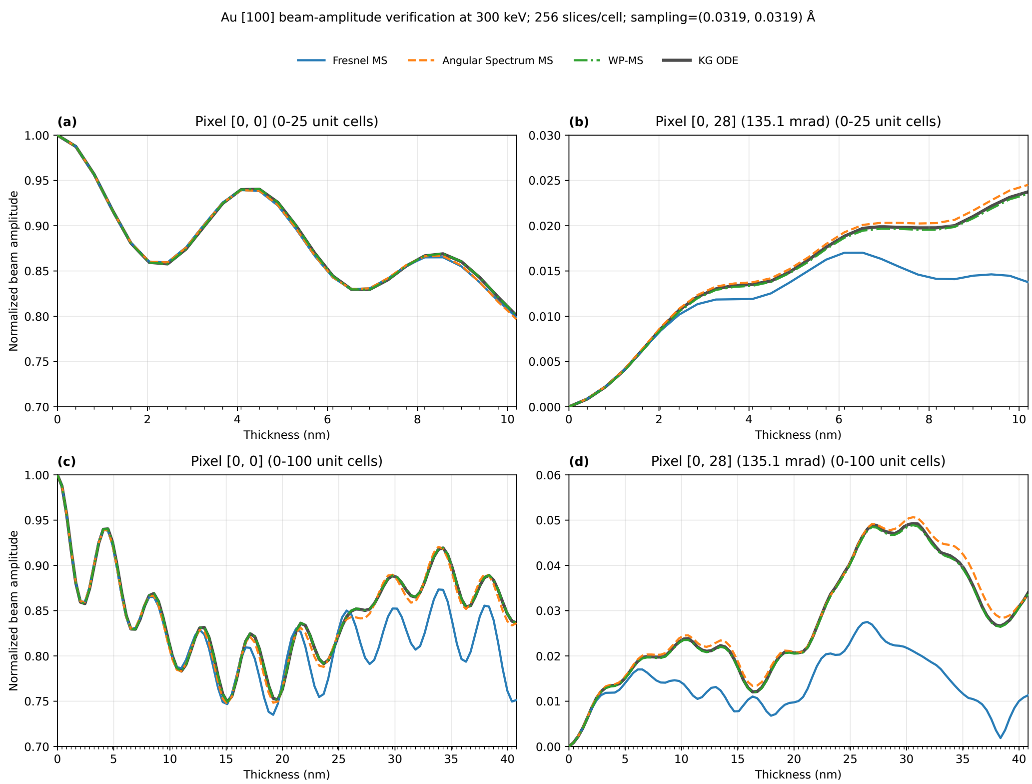

Figure 4: Fourier-component amplitudes for the Au [100] benchmark. The incident [0, 0] amplitude is normalised to unity. Panels (a) and (b) show the first 25 unit cells for the [0, 0] and [0, 28] components, respectively; panels (c) and (d) extend the same comparison to 100 unit cells. The [0, 28] component corresponds to approximately 135.1 mrad.

Figure 4 gives the corresponding Fourier-amplitude comparison at each unit-cell boundary. For the central [0, 0] component, all methods agree closely up to approximately 10 nm, or 25 unit cells. Beyond this thickness, the F-MS result begins to deviate from the other methods even on the on-axis pixel. By about 15 nm, high-angle diffracted beams have propagated far enough through the crystal to interfere back into the central beam, so an error in their accumulated phase also modifies the on-axis amplitude.

For the high-angle [0, 28] component at approximately 135 mrad, the F-MS solution diverges from the ODE reference after approximately 10 unit cells, or 4 nm, consistent with the findings of Rother and Scheerschmidt [3]. AS-MS tracks the ODE reference much more closely than F-MS, while WP-MS gives the smallest RMSE for both tracked Fourier components. The AS-MS–WP-MS difference is much smaller than the F-MS discrepancy over the thickness range considered here.

3.2. Si [111] CBED Peak Positions

Following the ODE comparison, we focus on the three FFT-based multislice methods: F-MS, AS-MS, and WP-MS. The first CBED test uses Si [111] at 100 keV with an aberration-free 8 mrad probe centred on a projected atomic column and a specimen thickness near 100 nm, following the example of the low-energy CBED comparison of Hillier et al. [4] but at a higher accelerating voltage. Starting from diamond Si with $a = 5.431$ Å, we constructed an orthogonal [111] cell, replicated it 12×7 laterally, and sampled it on a 1844×1863 grid with pixel sizes 0.049982 and 0.049985 Å. Each 9.4068 Å cell was divided into 32 slices, and 106 repeated cells gave a final thickness of approximately 100 nm.

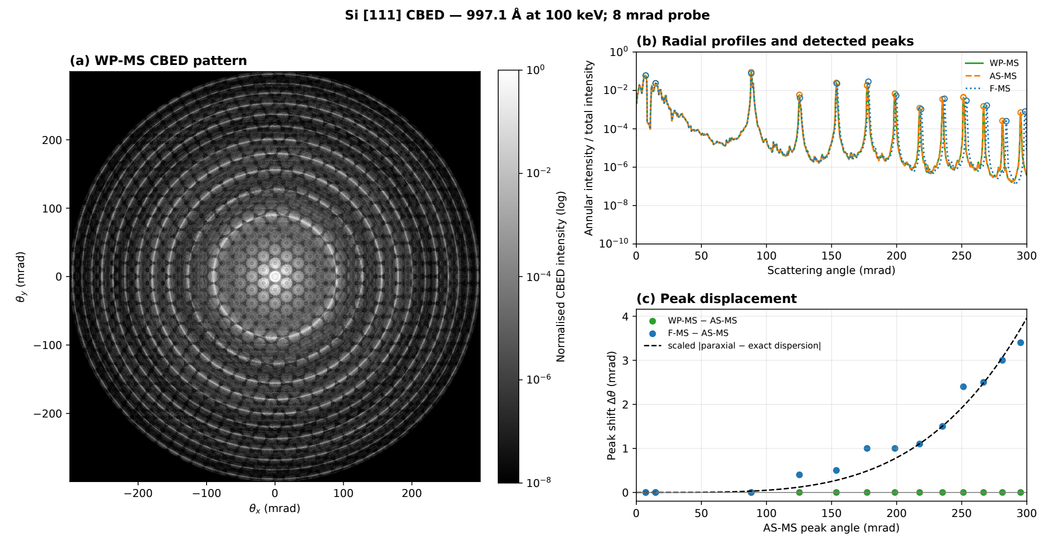

Figure 5: Si [111] CBED comparison at 100 keV, 8 mrad probe, and 99.7117 nm thickness. (a) The 128-bin WP-MS CBED pattern cropped to 300 mrad. (b) Radially integrated annular intensity for the three primary methods, normalised by the total pattern intensity, with detected diffraction peaks marked. (c) Peak displacements of F-MS and WP-MS relative to AS-MS, with a parabolic-versus-exact longitudinal-wavenumber error curve overlaid using one fitted scale factor. The F-MS–AS-MS shift has the same increasing trend as the dispersion-error curve, while the WP-MS–AS-MS shift remains near zero.

Figure 5 shows the WP-MS CBED pattern, the radial profiles with detected peaks, and the peak displacements relative to AS-MS. The F-MS peaks shift progressively with scattering angle. By contrast, AS-MS tracks WP-MS closely, with peak displacements near zero across the full 300 mrad range.

In Figure 5(c), the detected peak shifts are compared with a rescaled longitudinal-wavenumber error, namely the difference between the parabolic Fresnel dispersion and the exact spherical dispersion,

$$|\Delta\beta_z| = \frac{2\pi}{\lambda}\left|\sqrt{1-\lambda^2 q^2} - 1 + \frac{1}{2}\lambda^2 q^2\right|,$$

where $q = \sin\theta/\lambda$ is the transverse spatial frequency. This is the longitudinal phase error per unit propagation distance introduced when the exact spherical propagator is replaced by its Fresnel approximation; over a distance L, the corresponding phase error is $|\Delta\phi| = L|\Delta\beta_z|$. After fitting only the overall scale, the curve reproduces the increasing trend of the F-MS–AS-MS peak shifts. Because those two methods differ only in their free-space propagation kernel, this agreement is consistent with the displacement arising mainly from accumulated free-space propagation error between slices. In this Si case, the additional WP-MS correction from the potential-dependent wavelength inside the specimen is not clearly resolved in the peak positions.

3.3. Au [100] CBED Amplitude Differences

To examine the differences in a thick, high-Z specimen, we simulated convergent-beam electron diffraction from an Au [100] supercell at 200 keV using a 5 mrad aberration-free probe centred on a projected atomic column. The probe was propagated through 246 unit cells ($a = 4.08$ Å) to a specimen thickness of 100.368 nm, using 64 slices per unit cell ($\Delta z = 0.06375$ Å). To represent scattering angles beyond 300 mrad, the enlarged 9×9 gold cell was discretised on a 2048×2048 grid, giving a transverse pixel size of 0.01793 Å. We then propagated the wave using F-MS, AS-MS, and WP-MS, and compared the exit-surface CBED patterns obtained from the Fourier transform of each exit wave.

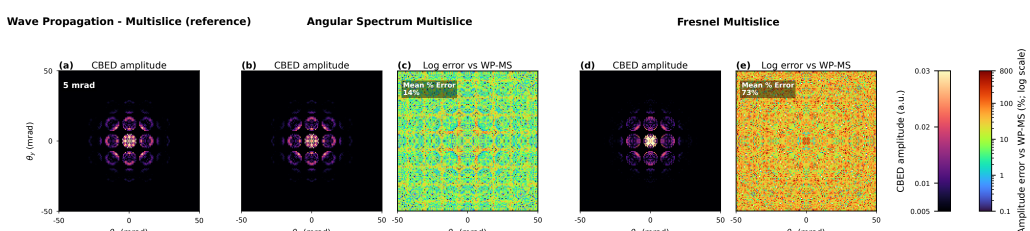

Figure 6: CBED exit-wave amplitude comparison for the Au [100] convergence-angle experiment. The top row shows the final diffraction amplitudes from (a) WP-MS, (b) AS-MS, and (c) F-MS. The lower row shows the pixel-wise amplitude errors of (d) AS-MS and (e) F-MS relative to WP-MS on a logarithmic percentage scale.

Figure 6 compares the final diffraction-plane amplitudes. The F-MS CBED amplitude differs visibly from WP-MS, especially in the higher-angle features of the pattern. AS-MS remains much closer to WP-MS, and the remaining AS-MS–WP-MS amplitude difference is small compared with the F-MS–WP-MS discrepancy.

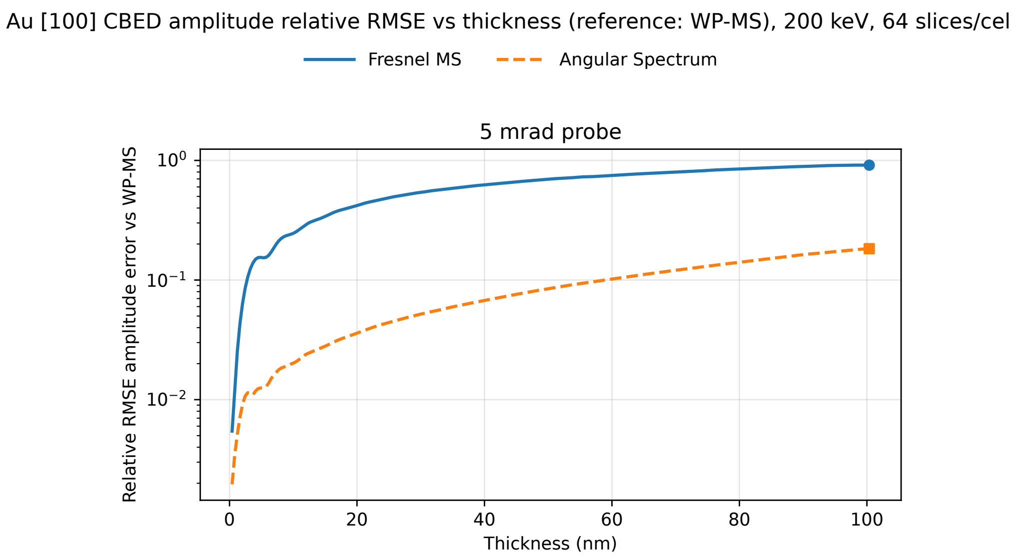

Figure 7: Thickness dependence of the Au [100] CBED-amplitude differences. Relative differences from the 64-bin WP-MS result are shown as a function of specimen thickness at 200 keV, using a 5 mrad probe and 64 slices per unit cell. Scattering angles up to 300 mrad are included.

Figure 7 shows how the CBED amplitude differs from the 64-bin WP-MS result as a function of thickness. After approximately 10 nm, the F-MS difference is 7.37–15.7 times the AS-MS difference. At 100.368 nm, the differences are 83.5% for F-MS and 11.3% for AS-MS. The AS-MS difference increases with thickness but remains smaller than the F-MS difference throughout the calculation. A residual AS-MS–WP-MS difference is also visible in the Au [100] exit-wave comparison of Figure 3, even at 40 nm.

4. Discussion

The three benchmarks identify a common source of error in conventional multislice under the high-angle, thick-specimen conditions tested here: the accumulated longitudinal phase error introduced by the parabolic Fresnel propagator. In the Au [100] ODE benchmark, the F-MS discrepancy appears first in the 135 mrad beam and only later in the central beam, after diffracted components have propagated far enough through the crystal to interfere back into it. In the Si [111] CBED calculation, the F-MS peak shifts increase with the parabolic-versus-exact dispersion error, while AS-MS remains close to WP-MS over the measured angular range. In the thick Au [100] CBED calculation, the F-MS amplitude difference becomes several times larger than the AS-MS difference beyond about 10 nm.

4.1. AS-MS as the minimal wide-angle correction

AS-MS removes most of this error by changing only the free-space propagation kernel. The transmission step, split-step ordering, FFT structure, and number of FFTs per slice are the same as in F-MS. In the spatial-frequency convention used here, the replacement is

$$\exp(-i\pi k_\perp^2 \Delta z/k_0) \quad \to \quad \exp\left(2\pi i \Delta z[\sqrt{k_0^2 - k_\perp^2} - k_0]\right).$$

The correction is small when all important Fourier components satisfy $k_\perp \ll k_0$, but it becomes significant when the simulation retains components with large $k_\perp/k_0 = \sin\theta$. Once those components are present, the error is not confined to high-angle beams because dynamical scattering can later mix them back into lower-angle amplitudes.

The AS-MS comparison is the main practical result. If a simulation grid already represents the high-angle scattering of interest, then replacing the Fresnel kernel by the angular-spectrum kernel removes the dominant wide-angle propagation error in these benchmarks without changing the basic multislice workflow.

4.2. Sampling and cost

The angular-spectrum propagation phase becomes steeper than the Fresnel phase at large transverse spatial frequency. For one propagation step,

$$\phi_{\mathrm{F}}(k_\perp) = -\pi \frac{\Delta z}{k_0} k_\perp^2, \qquad \phi_{\mathrm{ASM}}(k_\perp) = 2\pi \Delta z \left[\sqrt{k_0^2 - k_\perp^2} - k_0\right].$$

Comparing these gradients gives the relative phase-gradient factor

$$\frac{|d\phi_{\mathrm{ASM}}/dk_\perp|}{|d\phi_{\mathrm{F}}/dk_\perp|} = \frac{k_0}{\sqrt{k_0^2 - k_\perp^2}}.$$

This factor is only large when $k_\perp$ approaches k_0, where the propagating spectrum approaches the evanescent boundary. It can be written directly in terms of scattering angle because $k_\perp/k_0 = \sin\theta$:

$$\frac{k_0}{\sqrt{k_0^2 - k_\perp^2}} = \frac{1}{\sqrt{1 - \sin^2\theta}} = \frac{1}{\cos\theta}.$$

For scattering angles up to 300 mrad, this factor remains close to unity:

$$\frac{1}{\cos(300 \text{ mrad})} \approx 1.05,$$

so the phase-gradient factor increases by less than 5% at 300 mrad. The real-space sampling must also be fine enough to represent the required scattering angles. For square sampling Δx, angles up to $\theta_{\max}$ require $\Delta x \leq \lambda/[2\sin(\theta_{\max})]$. Components near the propagating–evanescent boundary require separate band-limiting and convergence checks because the angular-spectrum phase gradient diverges as $k_\perp \to k_0$. AS-MS has the same per-slice FFT count and asymptotic cost as F-MS, and both methods require sufficient transverse sampling for the scattering angles of interest.

4.3. WP-MS contribution

WP-MS makes a further correction by allowing the propagation kernel to depend on the local electrostatic refractive index, so the wavelength changes inside each step. In the electron-optics setting, this gives an automatic-differentiation-compatible one-way method that retains FFT-based propagation while incorporating a potential-dependent wavelength.

WP-MS has the smaller RMSE for both tracked Fourier components in the Au [100] ODE benchmark, although the size of its improvement over AS-MS is small; both methods remain much closer to the ODE reference than F-MS at 135 mrad. The residual AS-MS–WP-MS difference in the thick Au [100] CBED calculation contains both the additional local-wavelength correction and the WP-MS interpolation approximation. At the sampling and slice thickness used for that calculation, the largest sampled local refractive index is approximately 1.105, compared with 1 in vacuum, so the correction can accumulate over tens of nanometres. Under the high-energy conditions considered here, AS-MS captures most of the change from F-MS at the same per-slice FFT cost, whereas WP-MS produces a further increase in accuracy at a much higher cost.

To indicate the scale of this overhead in the present implementation, Table 2 reports propagation-only timings for the Au [100] CBED grid comparing F-MS, AS-MS, and WP-MS, whereby the WP-MS implementation is timed with 64, 128 and 256 refractive index bins in each slice. Each timed call propagated the same 2048×2048 probe through one 64-slice Au unit cell at 200 keV using double precision on an NVIDIA A100 80 GB GPU. The values are medians of ten calls after one warm-up call for each method; potential and probe construction, kernel construction, JAX compilation, and analysis were excluded. F-MS and AS-MS both required approximately 0.205 s per unit cell, whereas the WP-MS time increased approximately linearly with the number of refractive index bins. These values are implementation- and hardware-specific, but they show why the smaller AS-MS correction is the more practical default when the additional local-medium correction of WP-MS is not required.

Table 2: Propagation-only timing for one 64-slice Au [100] unit cell on a 2048×2048 grid using an NVIDIA A100 80 GB GPU. Relative times are normalised by the median AS-MS time. WP-MS times are reported for 64, 128, and 256 refractive index bins.

Method	**Median time (s)**	**Relative to AS-MS**
F-MS	0.205	1.00
AS-MS	0.205	1.00
WP-MS-64 bins	11.5	56.3
WP-MS-128 bins	22.8	111
WP-MS-256 bins	45.0	220

4.4. Consequences and limits

The silicon CBED calculation (Figure 5) shows that the same propagator error is not limited to the most severe high-Z case. Its CBED amplitude differences are smaller than in the thick Au [100] calculation, but the F-MS peak shifts still increase with the angular dispersion error. These high-angle peaks may be attenuated by thermal diffuse scattering in a room-temperature TEM experiment, and a conventional image simulation can hide part of the error if the transfer function or objective aperture suppresses the affected Fourier components. CBED comparisons and ptychographic forward models use diffraction-plane intensities more directly, so the same propagator error can become more visible.

In a ptychographic reconstruction, the forward model predicts detector intensities for each probe position. If the propagator gives the wrong high-angle amplitudes, peak positions, or accumulated phase through a thick specimen, the optimisation may compensate through the recovered potential. The calculation shows that the choice of propagator can change the diffraction-plane amplitudes before other experimental effects are added. For forward models that already use F-MS, most of the wide-angle correction can therefore be recovered by replacing the Fresnel kernel with the angular-spectrum kernel while keeping the same transmission step and per-slice FFT complexity.

The comparison here remains restricted to high-energy, elastic, forward-scattering calculations. At lower accelerating voltages, wide-angle propagation and backscattering become stronger, as does the refractive-index correction to the local wavelength. These conditions should therefore be tested separately [4–6]. The WP-MS implementation used here is one-way, but WPM also has bidirectional formulations [27–29], and adapting it to electron optics could provide a practical alternative to the FC-MS method of Chen and Van Dyck [8].

5. Conclusion

We adapted the wave propagation method developed in scalar optics to electron microscopy, giving a wave-propagation multislice (WP-MS) method that tracks high-angle propagation and local wavelength changes within an automatic-differentiation-compatible slice-based propagation framework. We then compared F-MS, AS-MS, and WP-MS in three complementary settings: an Au [100] ODE benchmark, a lower-Z Si [111] CBED peak-position test, and a thick Au [100] CBED amplitude test.

Across these tests, replacing the Fresnel kernel with the exact angular-spectrum kernel accounts for most of the improvement over paraxial Fresnel propagation. AS-MS keeps the same transmission step, FFT structure, and per-slice FFT cost as F-MS. For high-angle elastic forward simulations within the conditions studied here, AS-MS is therefore the most direct practical replacement for F-MS.

WP-MS includes an additional one-way local-medium correction and produces a numerically resolved difference from AS-MS under the high-energy conditions tested here. Its main role in this work is to bring the wave propagation method from scalar optics into electron optics and to provide an automatic-differentiation-compatible local-wavelength extension of angular-spectrum multislice. Future work should test WP-MS at lower accelerating voltages, compare it with bidirectional FC-MS, and evaluate it within ptychographic reconstructions, where systematic forward-model errors could bias the recovered potential.

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Data availability

The source code, input scripts, and numerical result archives that support the findings and reproduce the figures are publicly available in the WideAnglePropagation repository. The repository includes execution instructions and the numerical metadata recorded for each benchmark.

CRediT authorship contribution statement

David Landers: Conceptualisation, Methodology, Software, Validation, Investigation, Visualization, Writing – original draft. **Jean-Luc Rouvière:** Conceptualisation, Methodology, Writing – review and editing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI Codex to assist with language editing, code writing, and code review. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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